

Advanced Filtering & Business Question Answering

A Business Insight Sheet Built from Raw Order Data using Excel Conditional Formulas

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Role	Intern – Data Analytics
Task	Task 11 – Advanced Filtering & Business Question Answering
Dataset	Sample Superstore Dataset (120 records, 10 columns)
Tools Used	Microsoft Excel

1. Description

This task involved answering five targeted business questions directly from the raw order data using Excel's conditional and lookup formulas — rather than sorting or scanning the sheet manually. A supporting **Filter Raw Data** sheet was built alongside the raw table, using SUMIFS, COUNTIFS, AVERAGEIFS and a filtered/sorted helper view to isolate specific orders, regions, categories and segments on demand.

2. Objective

To practice using conditional aggregation formulas (SUMIFS, COUNTIFS, AVERAGEIFS) and sorting/filtering techniques to pull specific, decision-ready answers out of a raw transactional dataset, instead of relying on manual filtering or eyeballing the data for each question.

3. Business Questions & Answers

The table below lists the five business questions this task set out to answer, along with the result pulled from the 120-row raw order dataset.

#	Business Question	Answer
1	Which Region has the highest sales?	West — Rs. 7,15,050
2	Which Category is most profitable?	Office Supplies — Rs. 1,19,987.88 profit
3	Which orders resulted in a loss?	12 orders (out of 120)
4	What is the total sales of the Corporate segment?	Rs. 10,14,600
5	Which order(s) had the highest quantity?	Quantity = 15 units (11 orders tied, e.g. ORD-1009, ORD-1012, ORD-1032)

4. Supporting Breakdown

To see what sits behind the headline answers above, the raw data was also grouped by Region and by Category using SUMIFS.

4a. Region-wise Sales & Profit

Region	Orders	Sales (Rs.)	Profit (Rs.)
West	33	7,15,050	90,561.61
South	36	6,09,600	64,960.63
East	25	5,29,450	40,212.72
North	26	4,28,050	78,140.04

4b. Category-wise Sales, Profit & Margin

Category	Orders	Sales (Rs.)	Profit (Rs.)	Margin %
Office Supplies	49	10,68,200	1,19,987.88	11.2%
Furniture	40	6,98,800	75,759.75	10.8%
Electronics	31	5,15,150	78,127.37	15.2%

5. Write-up: What the Numbers Reveal

Taken together, the five answers tell a more specific story than the raw dataset does on its own.

- **West leads on sales, but North is the leaner region.** West brings in the highest regional sales at Rs. 7,15,050, yet North — with the lowest sales of the four regions at Rs. 4,28,050 — converts a much higher share of that into profit ($\approx 18\%$ margin) than West does ($\approx 13\%$ margin). Sales volume and profit efficiency don't move together region by region.
- **Office Supplies is the most profitable category despite the lowest per-order value.** It generates the highest total profit (Rs. 1,19,987.88) purely on volume — 49 of the 120 orders — even though its margin (11.2%) is not the best of the three categories. Electronics has the thinnest order count (31) but the healthiest margin (15.2%), while Furniture sits at the bottom on margin (10.8%).
- **Losses are rare but not negligible.** Only 12 of 120 orders (10%) were loss-making, meaning the business model is fundamentally sound — but a targeted look at those 12 orders (isolated in the Filter Raw Data sheet) is worth doing periodically, since a handful of them lose more than Rs. 1,500-Rs. 2,400 each.
- **Corporate is the anchor segment.** At Rs. 10,14,600, Corporate accounts for nearly 45% of total sales (Rs. 22,82,150) across all 120 orders — more than Home Office and Consumer combined are close to matching, but neither individually comes close to Corporate on its own.
- **High order quantity is common, not rare.** Eleven separate orders hit the maximum quantity of 15 units, spread across multiple categories and regions rather than being concentrated in one segment — so "largest order" isn't a single anomaly worth chasing, it's a recurring order-size pattern.

My Takeaway: Answering business questions straight from raw data with conditional formulas — rather than eyeballing a sorted list — makes the answers reproducible and audit-friendly. It also surfaces subtleties a single sorted view would hide, such as a region with lower sales still generating strong profit, or a category leading on profit purely through order volume rather than margin. The real value of SUMIFS/COUNTIFS here isn't speed — it's that the same five formulas will still give the right answer after the dataset grows well past 120 rows.